Distance from a point to a line

1 Write down the formula to find the perpendicular distance d from the point (x_1, y_1) to the line $Ax + By + C = 0$.

2 Find the perpendicular distance from the following points to the given line:

a Point $(0, 0)$, Line : $4x + 7y + 5 = 0$

b Point $(-9, -3)$, Line : $6x + 3y + 1 = 0$

c Point $(3, 5)$, Line : $4x + 3y + 1 = 0$

d Point $(4, -3)$, Line : $24x + 7y - 3 = 0$

e Point $(5, 4)$, Line : $5x - 12y = 0$

f Point $(-5, 3)$, Line : $3x - 2y - 5 = 0$

3 Find the perpendicular distance from the point (8, 7) to the line $9x - 9y - 0 = 0$.

4 Find the value of k with the following conditions:

a The perpendicular distance from the point $(-5, -1)$ to the line $-2x - 7y + k = 0$ is $\dfrac{10}{\sqrt{53}}$.

b The perpendicular distance from the point (($6, k$), to the line $6x + (-2y) + (-6) = 0$ is $\dfrac{40}{\sqrt{40}}$.

5 Consider the lines L_1: $x + 2y + 5 = 0$ and L_2: $x + by + 20 = 0$.

a Find the distance between L_1 and the origin.

b Solve for the value(s) of b such that L_1 is twice as far from the origin as L_2.

Geometric applications

6 Consider the circle represented by the equation $x^2 + y^2 = 0.81$.

a Find the centre of the circle.

b Find the radius of the circle.

c Find the perpendicular distance from the origin to the line $-4x + (-7y) + 8 = 0$.

d State the number of intersections between the line and circle. Explain your answer.

7 Consider the circle represented by the eq[redacted]n $(x-(-3))^2+(y-(-2))^2=4$.

a Find the centre of the circle.

b Find the radius of the circle.

c Find the perpendicular distance from the centre of the circle to the line $-3x+5y+(-5)=0$.

d State the number of intersections between the line and circle. Explain your answer.

8 The three points $A(-3, -4)$, $B(0, -10)$ and $C(-6, -7)$ form a triangle.

a Find the gradient of the line AC.

b Find the equation of the line AC in general form.

c Find the midpoint, D, of the line AC.

d Find the gradient of the line BD.

e Is BD perpendicular to AC?

f What type of triangle is ABC?

g Find the length of the line BD. Give your answer in simplest surd form.

h Find the length of the line AC. Give your answer in simplest surd form.

i Find the area of triangle ABC.

j Find the coordinates of the point E that would make $ABCE$ a parallelogram.

9 The points $A(5,7)$, $B(9,10)$, $C(6,4)$ and $D(2,1)$ are the vertices of a parallelogram.

a Determine the equation of the line going through AB.

b If AB is the length of the parallelogram, find the perpendicular height of the parallelogram.

c Determine the exact area of the parallelogram.